ASIAN HACKATHON FOR GREEN FUTURE 2026 — PROJECT PROPOSAL

FirmGrid — Sun-to-Wheels & Sun-to-Servers: the intelligence layer that turns Vietnam's wasted rooftop solar into firm clean power

TEAM NAME: [TEAM_NAME]

No.	Full Name	Role
1	[Member 1 — full name]	[Team lead / ML & optimisation]
2	[Member 2 — full name]	[Power systems & digital twin]
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1. PROJECT OVERVIEW

1.1 Project Title — FirmGrid — Sun-to-Wheels & Sun-to-Servers: the intelligence layer that turns Vietnam's wasted rooftop solar into firm clean power

1.2 Key challenge area — ☒ Renewable Energy and Low-Carbon Mobility ☐ Urban Air Quality and Climate Resilience ☐ Water Resources and Climate-Resilient Agriculture

1.3 Solution category — ☒ Software/Platform ☐ Mobile App/Web App ☐ Hardware/IoT ☐ Integrated Solution

1.4 Project Summary — Vietnam curtails rooftop solar at noon and burns coal after sunset, just as two demand waves arrive: electric two-wheelers pushed by Hanoi's Low-Emission Zone (from 1 July 2026) and data centres mandated to be ≥50% green by 2030. FirmGrid is an AI-orchestrated flexibility platform that makes the existing single-buyer grid intelligent instead of fighting it: GridMind forecasts, per neighbourhood transformer and 15-minute window, how much rooftop export is physically safe; a fair, explainable auction allocates that headroom among households; FlexMatch steers battery-swap and depot charging into the solar window (Tier 1 — Sun-to-Wheels). The same engine then aggregates orchestrated feeders and storage into firm, hour-matched 24/7 clean blocks sold to data centres via DPPA (Tier 2 — Sun-to-Servers). Beneficiaries: prosumer households, EV riders and swap operators, EVN distribution utilities, and Vietnam's digital economy.

2. CONTEXT AND PROBLEM STATEMENT

Vietnam built one of Asia's fastest solar booms (~18.8 GW by end-2024; PDP8 targets 46–73 GW by 2030), but generation peaks 10:00–14:00 while demand peaks 18:00–22:00, when coal — still over half of generation — carries the load. The binding constraint is local: 250–630 kVA neighbourhood transformers were sized for one-way flow, so on sunny late mornings EVN curtails rooftop output feeder-wide to protect equipment — blunt cuts, because operators cannot see which household could keep exporting safely. Reporting through April 2026 confirms these curtailments remain routine.

2026 adds a collision. Hanoi's LEZ starts on 1 July 2026 and the city plans to replace ~450,000 petrol motorbikes; VinFast alone operates ~4,500 battery-swap stations (target 45,000), with Selex and Honda scaling alongside — yet nothing coordinates when this flexible load charges, so by default it charges in the coal-heavy evening (~100 GWh/yr of new demand in Hanoi alone). Data-centre demand nearly doubles from ~735 MW (2025) to 1,330–1,543 MW by 2030 under the data-localisation law, with ≥50% mandated green. And on 26 June 2026 the rooftop surplus-sale ceiling was raised from 20% to 50% — more exportable energy pressing on the same constrained transformers, with no fair way to allocate it. Two visible failures — morning waste, evening dirt — share one root cause: no intelligence layer connects distributed solar, local grid state, and flexible demand.

3. EXISTING SOLUTIONS AND GAP ANALYSIS

- EVN status quo (Decree 58/2025 buyback + manual curtailment): trusted, universal — but reactive, feeder-wide, with no per-household granularity and no demand steering.
- Home batteries / residential VPPs (sonnen, Tesla): proven aggregation — but tens of millions of VND per home and no Vietnamese VPP regulatory category.
- Blockchain P2P trading (Powerledger — incl. the EVN CPC pilot; LO3): proved consumer appetite and transparent records — but it is a ledger: it records trades after the fact, has no predictive congestion model, and token settlement conflicts with Vietnamese payment law.
- SOLshare (Bangladesh): world-first P2P swarm grids and our benchmark for inclusive UX — but architected for off-grid DC nanogrids, not grid-tied urban feeders under a single buyer.

- DSO flexibility markets (Piclo Flex, GOPACS): the right congestion physics — but built for liberalised markets and professional aggregators, with no household UX.
- Uncoordinated smart charging (static time-of-use): no link to local surplus or transformer state.

The unresolved gap: no existing solution simultaneously (a) predicts distribution-level congestion before it happens, (b) allocates scarce export headroom fairly among ordinary households, (c) recruits mobility batteries as absorptive demand at solar noon, and (d) deploys inside Vietnam's single-buyer regulation without waiting for a new law. FirmGrid closes all four at once.

4. PROPOSED SOLUTION AND CORE FEATURES

FirmGrid replaces blunt curtailment with a predicted, market-cleared, minimal allocation — and gives the surplus somewhere useful to go. Five features, all demonstrated live in the working prototype (a physics-grounded digital twin of a real-topology 400 kVA Hanoi feeder: 30 PV homes, 25 non-PV homes, 3 C&I rooftops, 2 swap stations, 1 e-taxi depot, one simulated year at 15-minute resolution):

- F1 — GridMind Forecast: transformer breach probability and household surplus nowcast (gradient boosting; F1 = 0.80 on held-out days, feeder surplus MAE ≈ 2.4 kW in the twin — metrics shown on screen, honestly).
- F2 — HeadRoom Auction: a MILP clears every 15-minute window; accepted export never exceeds 90% of forecast headroom; rejection credits guarantee bounded maximum wait (measured max on the demo day: 1 window); every outcome explained in one sentence.
- F3 — Sun-to-Wheels FlexMatch: swap stations and depots are paid to pull charging into the 10:00–14:00 window — on the demo day this shifts ~300 kWh into the sun and recovers 95% of otherwise-curtailed energy (811 → 39 kWh wasted; zero limit breaches).
- F4 — One-Switch Prosumer App (roadmap: Zalo Mini App): default Auto-Sell, VND-first (“Hôm nay bạn kiếm được 12.400đ”), full bid→meter→payment transparency.
- F5 — TrustLedger + Sentinel: SHA-256 hash-chained audit of every event; bids are hard-capped at physically possible surplus, so an inflated 15 kW bid is blocked before money moves. No cryptocurrency — audit infrastructure, not tokens.

Tier 2 — Firm Block Studio: the same engine aggregates ~800 orchestrated transformers + 160 MWh storage into a 24/7 hour-matched block for a 20 MW data centre: 54% hourly CFE at a blended ~$84/MWh versus a ~$92/MWh grid tariff — the negotiation numbers of Vietnam's first firm-solar DPPA, computed live.

5. INNOVATION & COMPETITIVE ADVANTAGE

- Predict-then-allocate curtailment: “headroom” becomes a forecast product, market-cleared 15–60 minutes ahead — to our knowledge no deployed system offers household-granular, fairness-constrained allocation in a single-buyer market.
- Energy–mobility coupling at the feeder: swap networks become schedulable sinks for rooftop surplus — two national problems become each other's solution, in the launch city where both go live this quarter.
- Regulation-native design: households sell under Decree 58/2025 (cap 20%→50%, June 2026) with EVN as sole buyer; charging operators join as DPPA large consumers (Decree 57/2025); settlement is VND-only (Decree 52/2024). Ships without new law.
- Two-tier compounding: Tier 1's data flywheel (every onboarded feeder improves the forecasts) becomes Tier 2's product — firm 24/7 blocks for data centres — bridging Vietnam to its 2030–2035 nuclear baseload either way: if nuclear lands, FirmGrid prepared the flexible grid it plugs into; if it slips, FirmGrid is why the digital economy stayed clean.

Feasibility & development plan — Built during the hackathon: digital twin, GridMind, auction, FlexMatch, TrustLedger/Sentinel and Firm Block Studio, in a fully offline prototype with judge-in-the-loop controls (storm-front slider, fraud injection). Post-hackathon: Q3 2026 shadow pilot on one EVNHANOI feeder; Q4 2026 hardware-in-the-loop with ~20 volunteer homes via inverter-cloud APIs; H1 2027 regulatory-sandbox pilot with real money (200 households, 10 stations, EVN as counterparty); H2 2027 city scale (1,000 constrained transformers) plus the first firm-block DPPA pilot with a data centre; 2028 second city; 2029+ ASEAN (Thailand, Indonesia, the Philippines sit 3–5 years behind on the same curve). Risk honesty: if sandbox regulation lags, predictive minimal curtailment is a pure operations tool EVN can adopt unilaterally; market layers activate as regulation permits.

6. TARGET GROUPS & POTENTIAL IMPACT

Users: prosumer households (<100 kW rooftop, Decree 58/2025); swap-station and charging operators (VinFast / Selex / Honda ecosystems, e-taxi depots); EVN distribution companies (EVNHANOI first); EV riders (indirect); data centres and C&I rooftops via DPPA in Tier 2.

- Environmental: one constrained 400 kVA transformer strands ~24 MWh/yr; FirmGrid recovers 60–80% → 15–19 MWh. A 1,000-transformer Hanoi rollout recovers 15–19 GWh/yr ≈ 10,000–13,000 t CO_2 (official 2024 grid factor 0.681 tCO_2/MWh); a 20,000-transformer national scenario ≈ 200,000–260,000 t CO_2/yr. Steering 20–30% of the LEZ's ~100 GWh/yr charging wave into the solar window avoids a further 16,000–27,000 t CO_2/yr — about 1 kg CO_2 per swap, printed on the rider's receipt.
- Economic: households earn 0.6–1.0 million VND/yr per 5 kWp from energy that is thrown away today; stations save ~27–44 million VND/yr each by charging in the solar window; EVN defers transformer reinforcement measured in hundreds of millions of VND per site.
- Social: fairness bound guarantees small households their turn; every dong traceable from meter to payout; SDG 7, 9, 11, 13.

Committed KPIs: curtailed-kWh avoided; % of swap energy in the solar window; median household payout; transformer peak-loading delta; fairness index; fraud precision/recall; settlement latency; Tier-2 hourly CFE score.

7. DESCRIPTION OF TECHNOLOGIES APPLIED

A. Proposed technologies — Prototype (working today): Python, pandas/NumPy, scikit-learn gradient boosting, PuLP/CBC MILP auction, SHA-256 hash-chained ledger, Streamlit + Plotly dashboard; fully offline on one laptop. Scale-up roadmap: XGBoost/LSTM forecasting, OR-Tools, FastAPI, TimescaleDB, Redis Streams, MQTT, Zalo Mini App + React Native, Next.js + Mapbox operator console, optional permissioned Hyperledger Fabric, VNPay/MoMo sandbox payouts.

B. System architecture — a modular monolith in four planes (Data → Intelligence → Market → Experience) over event streams, splitting into services only under load. One 15-minute cycle: telemetry lands → GridMind publishes headroom h(t) and surplus $\hat{s}_i(t)$ → Auto-Sell agents and stations bid → MILP clears (Σ accepted $\leq 0.9 \cdot h(t)$, fairness bounds) → dispatch set-points → meter reconciliation → Sentinel anomaly screen → VND settlement → Merkle root sealed. Human operator override outranks every automated decision.

C. Data & infrastructure — Prototype: seeded synthetic twin calibrated to Hanoi climatology (Open-Meteo/PVGIS ranges) and Vietnamese load-research shapes; every assumption adjustable by judges. Pilot: EVN smart meters where deployed, inverter-cloud APIs elsewhere; partner station logs under MoU; personal data minimised, processed in-country. Infrastructure: demo = one laptop; cloud mirror ≈ US$40/month; pilot ≈ US$300–500/month.

8. REFERENCES

- Decree 58/2025/NĐ-CP (renewable energy development; rooftop surplus sale), as amended June 2026 (surplus-sale cap 20% → 50%); EVN Decision 429/QĐ-EVN (2025).
- Decree 57/2025/NĐ-CP on the DPPA mechanism (EV charging providers as large consumers); Decree 52/2024/NĐ-CP on non-cash payment.
- Decision 768/QĐ-TTg (April 2025), revised PDP8: solar 46–73 GW and 10–16 GW battery storage by 2030.
- SolarQuarter, “Vietnam Faces Rooftop Solar Curtailment Amid Grid Constraints” (13 Apr 2026); Norton Rose Fulbright, Vietnam Power Sector Snapshot (2025).
- Directive 20/CT-TTg (2025) and Hanoi People's Council LEZ resolution (Nov 2025); F&L Asia, “Hanoi to subsidise 450,000 motorbike switch” (2025).
- Vietnam.vn (Jan 2026): VinFast ≈4,500 swap stations, 45,000 target; Honda swap rollout from Apr 2026; Selex Motors shared-network materials.
- Ministry of Agriculture and Environment: official 2024 grid emission factor 0.681 tCO_2/MWh.
- IRENA (2025): firm solar+storage at USD 54–82/MWh in prime regions.
- Data-centre market: ~735 MW (2025) → 1,330–1,543 MW (2030) demand forecasts (Mordor Intelligence & industry reporting); Decision 2161 and the Jan-2026 data-localisation law.
- UNFCCC on SOLshare; Powerledger project materials (incl. EVN CPC pilot); Thurner et al., pandapower, IEEE TPS (2018); Chen & Guestrin, XGBoost, KDD (2016).